individual and all his future unborn brothers and sisters, with the mother taking the part of her future unborn children. More directly there may be competition between contemporary rivals for her investment, between litter mates or nest mates. Here, once again, the mother will normally be anxious to see fair play.

Many baby birds are fed in the nest by their parents. They all gape and scream, and the parent drops a worm or other morsel in the open mouth of one of them. The loudness with which each baby screams is, ideally, proportional to how hungry he is. Therefore, if the parent always gives the food to the loudest screamer, they should all tend to get their fair share, since when one has had enough he will not scream so loudly. At least that is what would happen in the best of all possible worlds, if individuals did not cheat. But in the light of our selfish gene concept we must expect that individuals *will* cheat, *will* tell lies about how hungry they are. This will escalate, apparently rather pointlessly because it might seem that if they are all lying by screaming too loudly, this level of loudness will become the norm, and will cease, in effect, to be a lie. However, it cannot de-escalate, because any individual who takes the first step in decreasing the loudness of his scream will be penalized by being fed less, and is more likely to starve. Baby bird screams do not become infinitely loud, because of other considerations. For example, loud screams tend to attract predators, and they use up energy.

Sometimes, as we have seen, one member of a litter is a runt, much smaller than the rest. He is unable to fight for food as strongly as the rest, and runts often die. We have considered the conditions under which it would actually pay a mother to let a runt die. We might suppose intuitively that the runt himself should go on struggling to the last, but the theory does not necessarily predict this. As soon as a runt becomes so small and weak that his expectation of life is reduced to the point where benefit to him due to parental investment is less than half the benefit that the same investment could potentially confer on the other babies, the runt should die gracefully and willingly. He can benefit his genes most by doing so. That is to say, a gene that gives the instruction ‘Body, if you are very much smaller than your litter-mates, give up the struggle and die’ could be successful in the gene pool, because it has a 50 per cent chance of being in the body of each brother and sister saved, and its chances of surviving in the body of the runt are very small anyway. There should be a point of no return in the career of a runt. Before he